AUTONOMOUS PRODUCTION CHOKE CONTROLLER

Problem Statement ID - Honeywell Hackathon, Problem 3A

Problem Statement Title - Autonomous Production Choke Controller for a Single Naturally Flowing Oil Well

Theme - Industrial Process Control / Oil & Gas Automation

PS Category - Software

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PROCESS UNDERSTANDING & MODEL

Step test: 120 hr open-loop, choke 30->40->55->45->65 %, Ts = 1 hr

Response is first-order (time constant ~5 hr on oil rate), mild noise

Opening choke RAISES oil rate but LOWERS WHP / FLP / BHP

=> the binding safety constraints are LOWER pressure bounds

Model assumptions: first-order + linear; single naturally flowing well

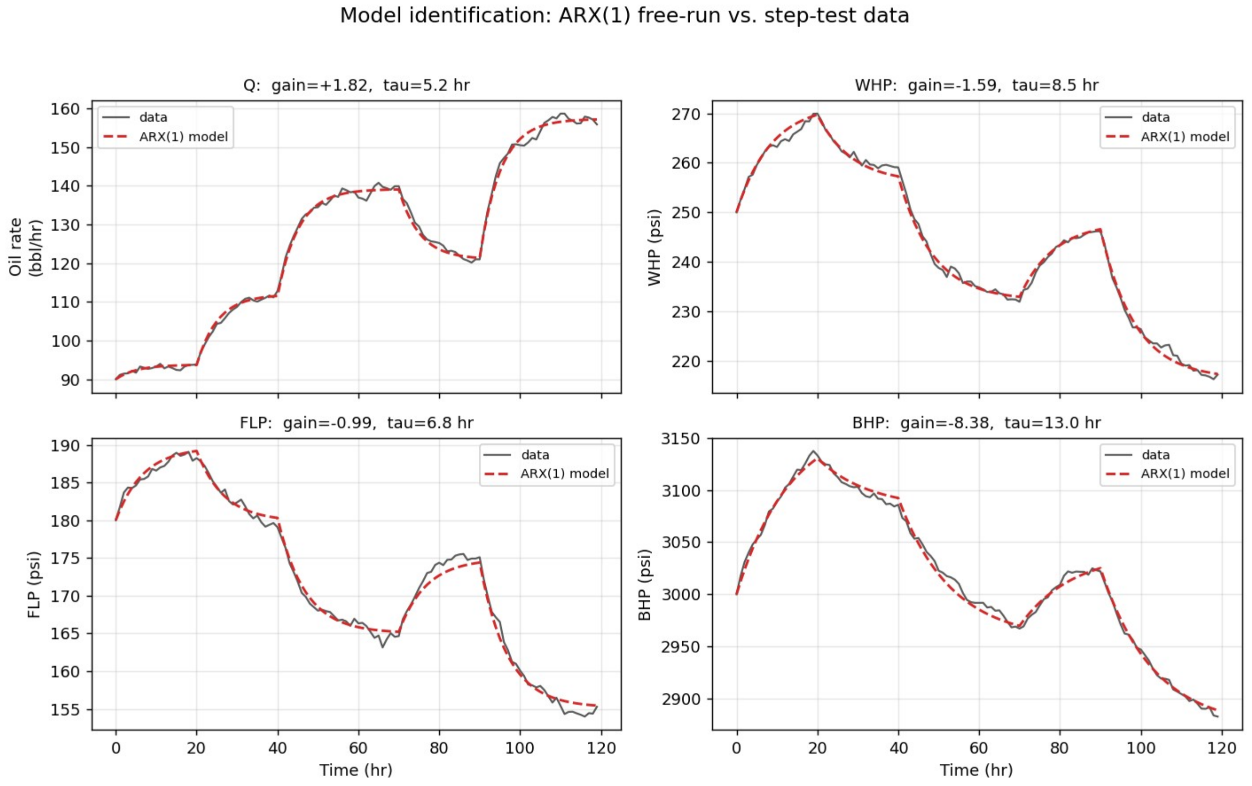
(no gas lift/ESP, constant reservoir & GOR); envelope limits documented

Model: ARX(1) per output by least squares

y[k+1] = a.y[k] + (1-a)(b0 + b1.u[k]); free-run fit 1.5-2.4 % of span

Identified steady-state gains (per % choke):

Q +1.82 bbl/hr   WHP -1.59   FLP -0.99   BHP -8.38 psi



CONTROL STRATEGY

Simplified MPC by brute-force candidate search (per brief)

Each hour, enumerate choke +/- up to 5 % (ramp-feasible, fine grid)

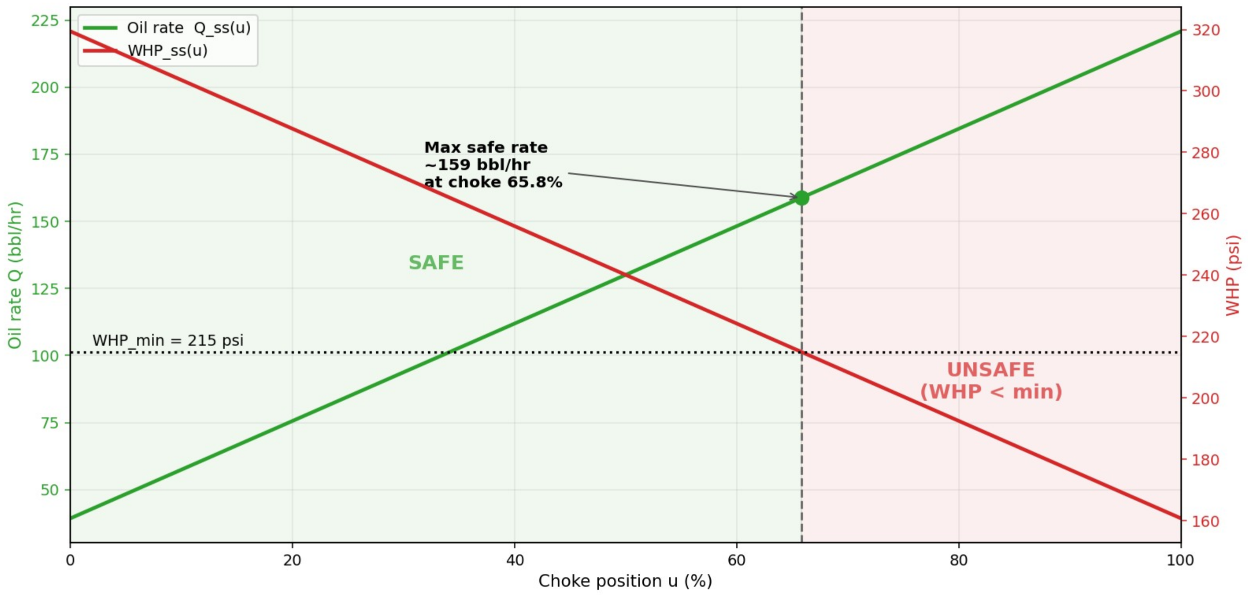
Predict each candidate's pressures over a settling horizon + steady state

REJECT any move predicted to breach WHP/FLP/BHP + safety margin

Among safe moves, minimise (Q_ss - target)^2 + lambda.(delta choke)^2

Infeasible target => same cost settles at the highest safe rate

Monotonic first-order response: steady-state feasibility proves the whole trajectory safe -> safe-by-construction, 0 violations



RESULTS - TARGET TRACKING (A & B)

Scenario A - Startup -> 130 bbl/hr

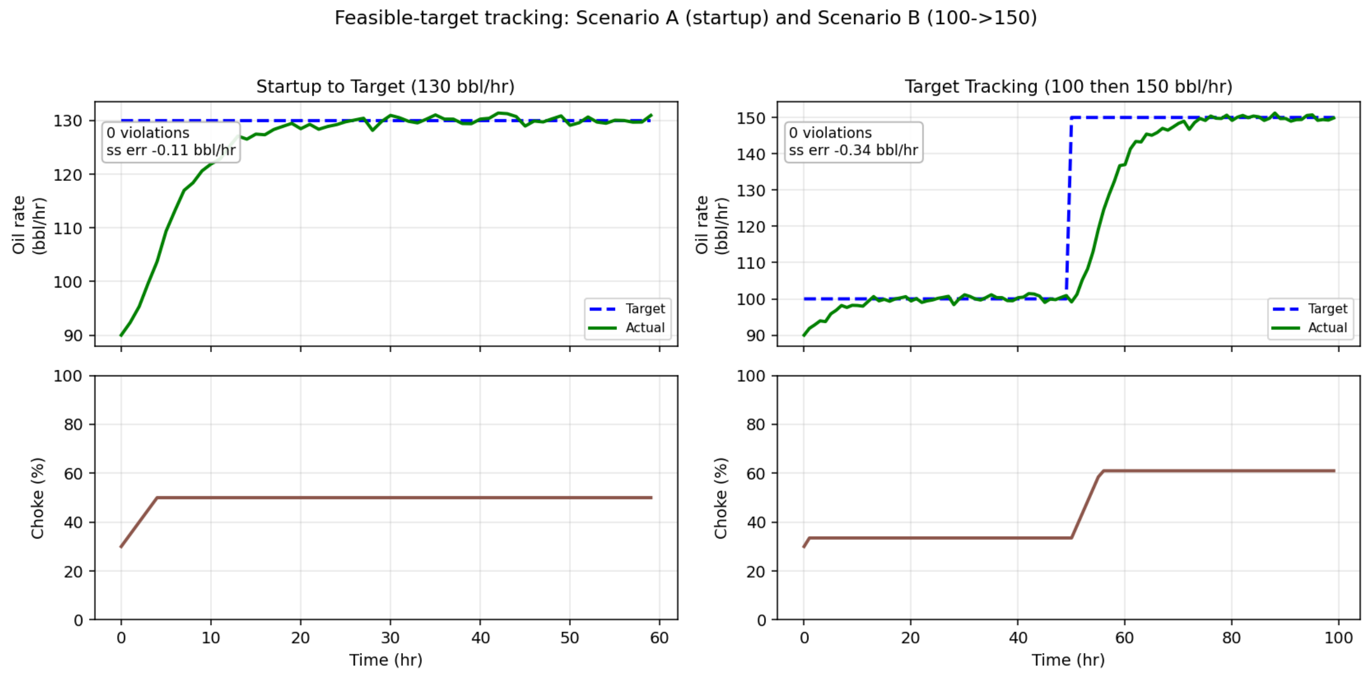
settles 129.9 bbl/hr, steady-state error -0.11, 0 violations

Scenario B - Target step 100 -> 150 bbl/hr

settles 149.7 bbl/hr, steady-state error -0.34, 0 violations

Pressures stay inside the envelope throughout

Choke moves are smooth and respect the +/-5 %/step ramp limit



SAFETY PERFORMANCE & LESSONS LEARNED

Scenario C - infeasible 185 bbl/hr: controller REFUSES,

settles at max safe 154.9 bbl/hr with 0 violations

Robust: 0 violations under +/-20% plant mismatch, unmeasured disturbances and 2x noise (22-case stress study, robustness.py)

Lessons learned:

Binding constraint is a lower pressure bound, not an upper rate cap

Monotonicity makes a cheap brute-force MPC provably safe

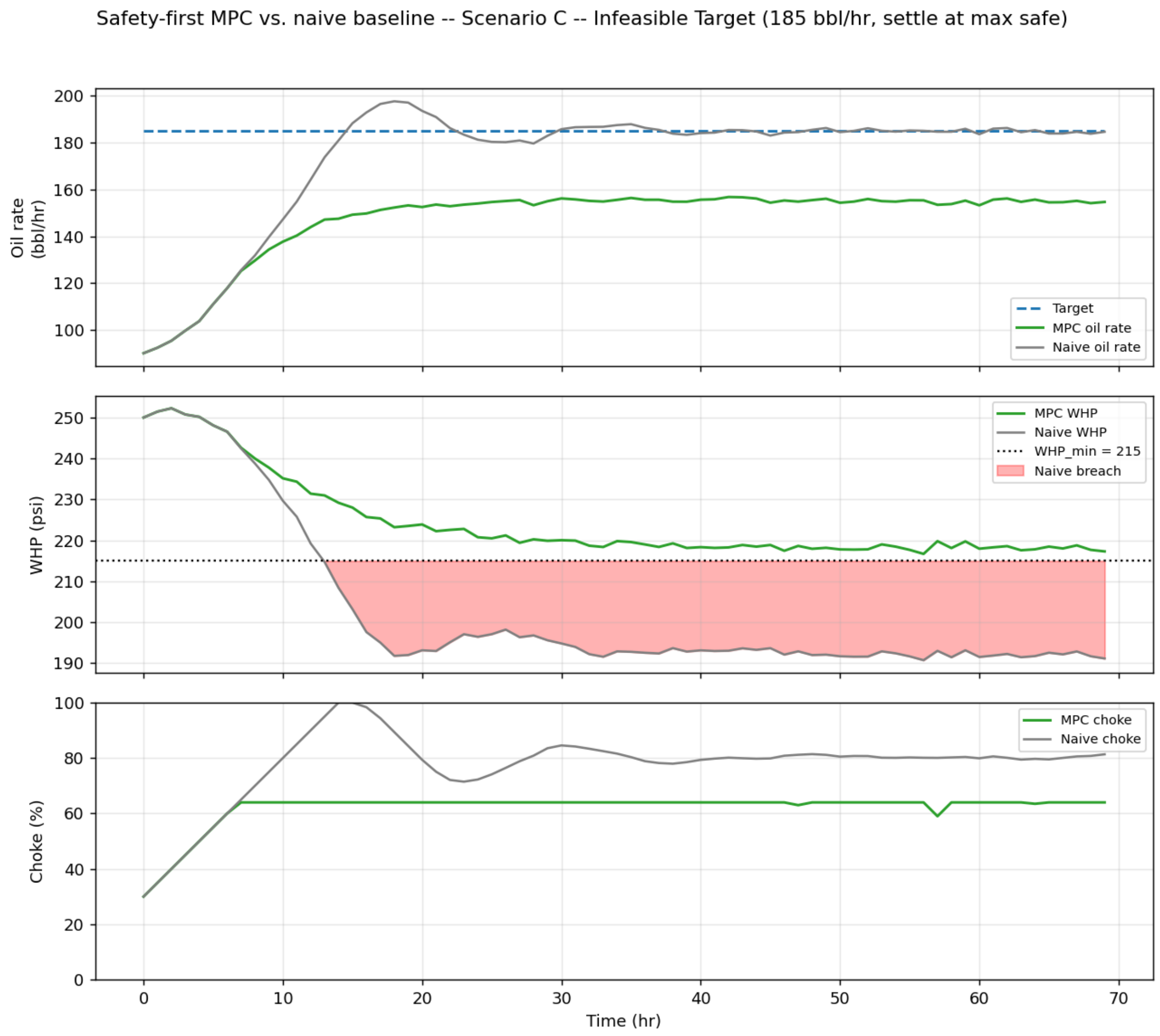
Model error costs production, never safety - margin is the dial

0

MPC violations (A/B/C)

168

naive-baseline breaches on C



ARTIFACTS & REFERENCES

Code (flat Python modules at repo root):

dataio, model_id, envelope, simulator, controller, plotting, scenarios, main

Executed Jupyter notebook + technical report.md + results/ (plots, metrics.json)

Live dashboard: streamlit run app.py (scenario presets, per-step

MPC diagnostics: candidates rejected as unsafe, WHP margin)

Robustness: python robustness.py (22 stress cases: noise, +/-20%

mismatch, unmeasured disturbances)

Run: pip install -r requirements.txt && python main.py

main.py asserts 0 envelope violations across A/B/C

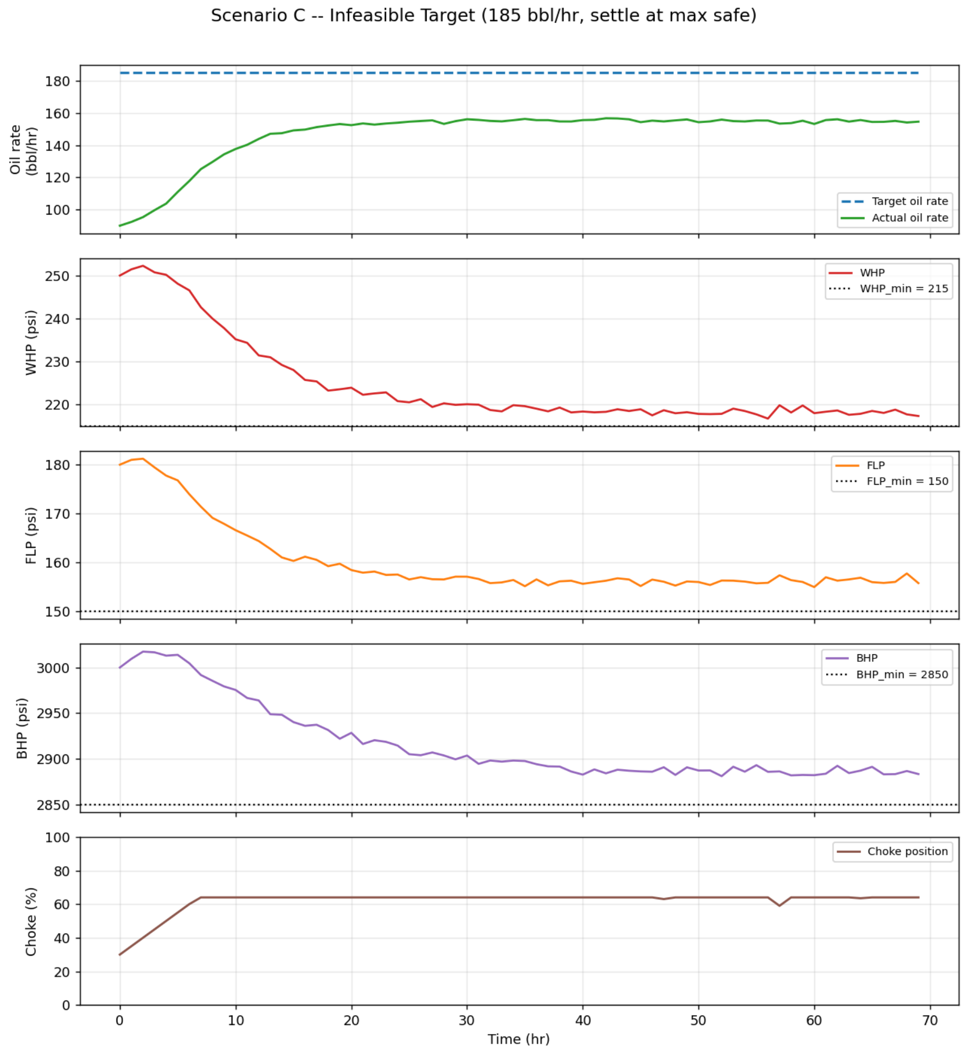
Simulator stand-in shares the official .step(choke)->(Q,WHP,FLP,BHP)

signature -> the real event simulator drops in with a one-line change

Repo: github.com/Divyansh2602/autonomous-choke-controller

References: least-squares system identification (Ljung); simplified

MPC by brute-force candidate evaluation (Honeywell brief)



Required 6-trend plot (Scenario C shown)